

Deep Fake Video Detection Using Attention-Based Convolutional Neural Network

1st S. Harivallabha Sai

CVR College of Engineering
(Affiliated to JNTUH)
Hyderabad, India
vallabha1243@gmail.com

2nd G. Shruthi

CVR College of Engineering
(Affiliated to JNTUH)
Hyderabad, India
guvalashruthi@gmail.com

3rd D. Sohan

CVR College of Engineering
(Affiliated to JNTUH)
Hyderabad, India
dsohan9999@gmail.com

4th Ch.Bhavani

CVR College of Engineering
(Affiliated to JNTUH)
Hyderabad, India
chbhavani99@gmail.com

Abstract—The increasing realism of deepfake videos complicates the verification of digital media. This paper describes a deepfake video detection framework that combines a convolutional neural network (CNN) with a Convolutional Block Attention Module (CBAM) to emphasize informative facial features. The proposed pipeline samples video frames, detects and crops faces, normalizes facial images, and classifies them as authentic or manipulated. Grad-CAM is included to visualize image regions that influence predictions. The evaluation protocol is designed to report accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices; compare the attention model with an equivalent CNN without attention; test cross-dataset generalization from FaceForensics++ to Celeb-DF; and assess performance after compression and resolution reduction. A FastAPI service and web frontend are specified for video submission and result presentation. Numerical experimental results and implementation-specific model settings were not available for this draft, so no performance claims are made.

Index Terms—deepfake detection, attention mechanisms, CBAM, video forensics, media authentication

I. INTRODUCTION

The growing quality and accessibility of deep-learning-based face manipulation have made it increasingly difficult to distinguish authentic videos from manipulated content by visual inspection alone. Deepfakes can be misused for misinformation, impersonation fraud, and non-consensual content, while uncertainty about authenticity can also undermine trust in genuine media. These risks motivate practical detection methods for digital forensics, journalism, cybersecurity, and content moderation.

CNN-based detectors learn visual representations from facial images and video frames. However, subtle manipulation traces may be concentrated around facial regions such as the eyes, mouth, and blending boundaries. A conventional CNN does not explicitly encode which spatial features should receive greater emphasis. In addition, measured performance depends on manipulation types, dataset composition, split design, compression, and evaluation protocol. Strong results on one benchmark alone do not establish generalization to other datasets or real-world video conditions.

This paper presents a framework for deepfake video detection using a CNN augmented with a Convolutional Block Attention Module (CBAM). Its intended pipeline extracts frames from a submitted video, detects and crops faces, normalizes the

resulting images, and classifies facial content. The framework also includes Grad-CAM visualizations and a web application architecture with a FastAPI backend. Its evaluation plan covers in-domain metrics, a no-attention ablation, cross-dataset testing, and degraded-video conditions. The contributions of this work are:

- A video-level detection pipeline design that combines facial preprocessing with CNN feature extraction and CBAM attention.
- An evaluation protocol for independently examining attention, cross-dataset transfer from FaceForensics++ to Celeb-DF, and robustness to compression and reduced resolution.
- An explainability and deployment design that pairs Grad-CAM visualizations with a FastAPI prediction service and a web-based results interface.

These contributions describe the proposed framework and evaluation design. The available project material does not include completed experimental results or enough implementation details to reproduce a specific trained model; accordingly, this paper does not claim measured improvements.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the proposed method. Section IV presents the experimental setup, Section V reports results, and Section VI concludes the paper.

II. RELATED WORK

A. Deepfake Datasets

B. Benchmark Datasets

FaceForensics++ provides manipulated facial videos and a benchmark for evaluating forgery-detection methods [?]. Celeb-DF was introduced as a challenging dataset containing higher-quality synthesized videos, supporting evaluation under more realistic manipulation conditions [?]. Dataset choice and partitioning are consequential: shared identities, source videos, or related frames across training and test partitions can inflate estimates of generalization. Cross-dataset evaluation is therefore useful for measuring domain shift, provided label mapping and preprocessing are documented.

C. CNN-Based Detection

CNN-based systems learn visual patterns that may distinguish manipulated from authentic media. Xception uses depth-wise separable convolutions and has been applied to image classification [?]. MesoNet was designed as a compact network for facial video forgery detection [?]. These architectures motivate CNN feature extraction for facial forgery analysis. Comparisons remain meaningful only when datasets, splits, preprocessing, and evaluation units are consistently reported.

D. Positioning of This Study

E. Attention and Explainability

The Convolutional Block Attention Module sequentially applies channel and spatial attention to intermediate CNN features, allowing the network to adaptively emphasize informative feature maps and locations [?]. Grad-CAM uses gradients flowing into a convolutional layer to produce a coarse localization map that indicates image regions influential to a class prediction [?]. In a deepfake detection system, these methods can support attention-guided feature learning and visual inspection of predictions, respectively. A heatmap is an explanation of model sensitivity, however, and should not be interpreted as proof that a highlighted region contains a manipulation artifact.

F. Research Gap and Scope

Prior work establishes CNN-based detection and attention modules as relevant tools, while the benchmark datasets enable evaluation across manipulation sources. This project focuses on bringing several evaluation dimensions into one proposed application: a controlled comparison with an otherwise equivalent CNN without CBAM, cross-dataset testing using FaceForensics++ and Celeb-DF, robustness tests under compression and reduced resolution, and Grad-CAM visualization. This combination is the scope of the proposed framework, not a claim that each component is unprecedented. Its empirical value remains to be established through completed, consistently controlled experiments.

III. METHODOLOGY

A. System Overview

The proposed system accepts a video through a web interface and processes it through a FastAPI backend. The backend extracts frames, detects and crops facial regions, resizes and normalizes face images, and passes them to the classifier. The model produces frame-level predictions; a video-level prediction is then formed by aggregating those scores. The aggregation rule, frame-sampling rate, face detector, input dimensions, and behavior when no face is detected must be fixed and reported by the implementation. Data augmentation is applied during training only, with transformations selected to preserve the class label. The prediction, confidence score, and selected Grad-CAM visualization are returned to the frontend for display.

Add the workflow diagram as
figures/workflow.png

Fig. 1. Proposed video detection workflow from upload and facial preprocessing to classification and visual explanation.

Add the architecture diagram as
figures/architecture.png

Fig. 2. Conceptual CNN and CBAM architecture. The final diagram should reflect the selected implementation.

B. CNN Architecture

The detector is designed as a CNN backbone augmented with CBAM. For an intermediate feature tensor, CBAM first infers channel-wise weights and then spatial weights, refining the features before classification [?]. The project description identifies Xception and ResNet as candidate backbones but does not specify which one was selected, nor does it provide the classifier head, input dimensions, initialization, or parameter count. These details must be recorded for a reproducible implementation; no particular backbone or trained configuration is assumed here. The ablation model should use the same backbone, classifier, data, and training protocol, with CBAM as the only architectural difference. Fig. 2 is optional.

C. Training Objective

For binary classification, a common objective is binary cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)], \quad (1)$$

where N is the number of training examples, $y_i \in \{0, 1\}$ is the ground-truth label, and p_i is the predicted probability for class 1. Binary cross-entropy is the intended objective for binary classification; the final report should confirm the loss and any class weighting used by the training code.

D. Inference

For each detected face, the classifier returns a score for the manipulated class. A decision threshold converts this score to a frame-level label, and an aggregation rule combines frame scores into the video-level confidence and label. Neither the threshold nor the aggregation rule is specified in the supplied project description, so both are implementation details that must be stated before reporting video-level metrics. Grad-CAM is computed for a selected convolutional layer and overlaid on the corresponding face crop to visualize regions contributing to the prediction [?]. Heatmaps are explanatory visualizations and are not independent evidence of manipulation.

E. Application Architecture

The application is organized into a video-upload frontend, a FastAPI service, video and face preprocessing, model inference, Grad-CAM generation, and a result dashboard. The

service packages the predicted label, confidence score, and visualization for the frontend. This architecture describes the intended end-to-end workflow; deployment performance, supported file limits, and operational behavior should be reported after they are verified in the running application.

IV. EXPERIMENTAL SETUP

A. Dataset and Splits

FaceForensics++ is designated as the primary dataset for training and in-domain evaluation, with Celeb-DF reserved for cross-dataset testing. The supplied project description does not identify dataset versions, compression settings, sample counts, manipulation categories included, or actual partition counts. These must be reported from the data used in the final run. Splits should be performed at the source-video level, and, where applicable, identities should not overlap between training and test partitions. All sampled frames from a video must remain in the same partition to prevent leakage. Any filtering, class balancing, or label mapping must also be documented.

B. Implementation Details

The proposed software stack uses Python and PyTorch with torchvision for model and image-processing components, OpenCV for video decoding and frame extraction, and FastAPI for the prediction service. MediaPipe or MTCNN is identified as a possible face detector, but the selected implementation is not specified. The project description also names scikit-learn for metric computation and a web frontend for presenting results. Framework versions, exact hardware, backbone, input resolution, optimizer, learning rate, batch size, epochs, augmentation parameters, random seed, checkpoint-selection rule, and trained model weights were not supplied. These values are necessary for reproducing and interpreting an experimental result.

C. Evaluation Metrics

The evaluation protocol reports accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix, with manipulated video defined as the positive class. For a binary classifier, precision and recall are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively. Metrics should be computed at the video level using the aggregated video prediction, and this unit should be kept consistent across comparisons. ROC-AUC should use continuous prediction scores rather than thresholded labels.

D. Ablation, Generalization, and Robustness

The attention ablation compares the CBAM model with an otherwise equivalent CNN without attention, using identical splits, preprocessing, training budget, and evaluation metrics. Cross-dataset generalization is assessed by training on FaceForensics++ and testing on Celeb-DF without fitting

Add a results plot as
 figures/results.png

Fig. 3. Evaluation results; populate this figure only with measurements from completed experiments.

to the target test set. Robustness is assessed by evaluating copies of test videos after controlled compression and spatial downsampling; codec, compression quality, target resolution, and any resizing applied before inference must be recorded. No baseline or test result should be presented unless the corresponding experiment has been run under a documented protocol.

V. RESULTS AND DISCUSSION

The supplied project material describes the intended metrics and evaluation scenarios but does not include completed runs, prediction outputs, or verified numerical measurements. Consequently, this section does not report accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, or comparative gains. Such values cannot be inferred from the proposed architecture or evaluation plan.

The final evaluation should report in-domain results on the held-out FaceForensics++ test partition, the matched no-CBAM ablation, and cross-dataset results on Celeb-DF. Compression and reduced-resolution results should be presented alongside the unmodified test condition, with the degradation settings specified. The discussion should analyze false positives and false negatives, distinguish within-dataset performance from cross-dataset transfer, and limit conclusions to the tested data and conditions. Fig. ?? can present measured results once the experiments are complete.

A. Limitations

The present draft is limited by the absence of a confirmed backbone configuration, training details, data split counts, and experimental measurements. The proposed evaluation also covers only the named datasets and degradation settings; it cannot establish performance across all manipulation methods, populations, or distribution shifts. Compression and resolution changes may remove or alter visual cues, while face-detection errors and frame aggregation can affect video-level decisions. Demographic effects, calibration, and operational latency remain unmeasured in the available material. These are limitations of the evidence currently documented, not findings about the model's measured behavior.

VI. CONCLUSION

This paper describes a proposed deepfake video detection framework that combines facial frame preprocessing, CNN classification, and CBAM attention, with Grad-CAM visualizations and a FastAPI-backed web application architecture. Its evaluation plan includes a controlled no-attention ablation, training on FaceForensics++ with cross-dataset testing on Celeb-DF, and robustness assessment under compression and

reduced resolution. These elements address attention validation, domain shift, degraded inputs, and prediction interpretability as explicit evaluation goals.

The available project documentation does not provide a confirmed model configuration or completed experimental results. Therefore, no conclusion about detection accuracy, the benefit of CBAM, cross-dataset generalization, or robustness can yet be drawn. The next step is to finalize and document the implementation and data protocol, run the specified experiments, and report measured results with video-level metrics. Future extensions may include real-time stream analysis, audio-visual detection, mobile deployment, and training on larger and more diverse datasets.